

Per-trajectory Trotterization of spin dynamics under isotropic rotational diffusion: sampling the $SO(3)$ Wiener process and its white-noise limit

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Abstract

We implement “Route 1” for a single spin- $\frac{1}{2}$ with chemical-shift anisotropy (CSA) under isotropic rotational diffusion. Each trajectory samples the molecular orientation as a Brownian motion on $SO(3)$, and the spin propagator is Trotterized with the orientation held piecewise constant. We compare three increment samplers: the exact heat kernel, the Gaussian geometric-Euler step, and a curvature-corrected Gaussian. We show analytically and numerically that the ensemble-averaged scheme converges with weak order two when the increments are exact, and with order one for the naive Gaussian step. The corrected variance $2D_R\tau(1 - D_R\tau/6)$ restores order two. The piecewise-constant scheme needs $\tau \lesssim \tau_c$; its discrete-time spectral density is available in closed form. In the white-noise limit ($\tau_c \rightarrow 0$ at fixed $\Gamma = \delta^2\tau_c$) the orientation drops out of the problem. Route 1 then becomes a Trotterized Stratonovich SDE with Gaussian “kicks” whose covariances are the Redfield spectral densities, and a frequency-resolved variant of it reproduces Redfield theory for all $\omega_0\tau_c$. For liquid-state NMR parameters this is the only practical form of Route 1. Section 5 extends both schemes to a dipolar-coupled heteronuclear pair with CSA on both spins. All interactions are linear images of one rank-2 orientation field. Route 1 therefore obtains DD/CSA cross-correlation (longitudinal $I_z \rightarrow 2I_zS_z$ transfer, TROSY) by sharing one $R(t)$, and the kick scheme obtains it by sharing one Gaussian 5×5 integrated-field matrix per Zeeman frequency. Route 1 also exposes odd-order corrections from the non-Gaussian statistics of $D^{(2)}(R)$, which Gaussian kicks cannot capture and which are negligible in liquids.

1 Model

Let $R(t) \in SO(3)$ map the CSA principal-axis frame (PAS) to the lab frame. With $\hbar = 1$, the lab-frame Hamiltonian is

$$H(R) = \omega_0 I_z + \mathbf{b}(R) \cdot \mathbf{I}, \quad \mathbf{b}(R) = \delta R A R^T \hat{\mathbf{z}}, \quad A = \text{diag}\left(-\frac{1+\eta}{3}, -\frac{1-\eta}{3}, \frac{2}{3}\right), \quad (1)$$

where $\delta = \omega_0 \Delta\sigma$ is the anisotropy in angular-frequency units and η is the asymmetry. The isotropic shift is absorbed into ω_0 . For an axial tensor ($\eta = 0$) with unique axis $\mathbf{n} = R\hat{\mathbf{z}}$,

$$\mathbf{b} = \delta(n_z \mathbf{n} - \frac{1}{3} \hat{\mathbf{z}}), \quad b_z = \frac{2}{3} \delta P_2(n_z), \quad b_{x,y} = \delta n_z n_{x,y}. \quad (2)$$

All components of $\mathbf{b}$ are pure rank-2 functions of R . Under isotropic diffusion they are therefore exponentially correlated with the same time $\tau_c = 1/(6D_R)$, and their equal-time covariance is diagonal:

$$C_{ij} = \langle b_i b_j \rangle = \delta^2 \text{diag}\left(\frac{1}{15}, \frac{1}{15}, \frac{4}{45}\right)_{ij}, \quad \langle b_i(t) b_j(0) \rangle = C_{ij} e^{-|t|/\tau_c}. \quad (3)$$

Redfield theory with $J(\omega) = \frac{2}{5}\tau_c/(1 + \omega^2\tau_c^2)$ gives

$$R_1 = \frac{1}{3} \delta^2 J(\omega_0) = \frac{2}{15} \frac{\delta^2 \tau_c}{1 + \omega_0^2 \tau_c^2}, \quad R_2 = \frac{1}{18} \delta^2 [4J(0) + 3J(\omega_0)] = \frac{4}{45} \delta^2 \tau_c + \frac{1}{2} R_1. \quad (4)$$

For a spin- $\frac{1}{2}$, $e^{-i\tau \mathbf{h} \cdot \mathbf{I}}$ rotates the Bloch vector $\mathbf{M}$ about $\mathbf{h}$ by the angle $|\mathbf{h}|\tau$. The code therefore propagates Bloch vectors with Rodrigues’ formula, which is exact and cheap to vectorize over trajectories.

2 Rotational diffusion as a Wiener process on $SO(3)$

Isotropic rotational diffusion is the Brownian motion on $SO(3)$ driven by a white-noise angular velocity. In Stratonovich form,

$$dR = \sqrt{2D_R} R \sum_{a=1}^3 L_a \circ dW_a, \quad (L_a)_{bc} = -\epsilon_{abc}, \quad \langle dW_a dW_b \rangle = \delta_{ab} dt, \quad (5)$$

with generator $D_R \nabla_{SO(3)}^2$. Its eigenfunctions are the Wigner matrices, $\nabla^2 D_{mk}^l = -l(l+1)D_{mk}^l$. Because the diffusion is isotropic, left and right increments are equivalent in law. We use body-frame increments,

$$R_{k+1} = R_k Q_k, \quad Q_k \text{ i.i.d. isotropic.} \quad (6)$$

Any isotropic increment law is a class function (it depends only on the rotation angle ω). By Schur's lemma it acts on each irrep as a scalar,

$$\mathbb{E}[D^l(Q)] = f_l \mathbb{K}_{2l+1}, \quad f_l = \frac{\mathbb{E}[\chi_l(\omega)]}{2l+1}, \quad \chi_l(\omega) = \frac{\sin((l+\frac{1}{2})\omega)}{\sin(\omega/2)}. \quad (7)$$

The exact propagator requires $f_l = e^{-l(l+1)D_R\tau}$ for every l . We compare three samplers.

(i) Exact heat kernel. Draw the axis uniformly on S^2 and the angle from

$$p(\omega) = \frac{2}{\pi} \sin^2 \frac{\omega}{2} \sum_{l \geq 0} (2l+1) e^{-l(l+1)D_R\tau} \chi_l(\omega), \quad \omega \in [0, \pi]. \quad (8)$$

The CDF is available in closed form, since $\int_0^\omega \sin^2(\frac{w}{2}) \chi_l(w) dw = \frac{1}{2} \left[\frac{\sin l\omega}{l} - \frac{\sin(l+1)\omega}{l+1} \right]$ (and $\frac{1}{2}(\omega - \sin \omega)$ for $l=0$). We tabulate it once per τ and invert by interpolation. The series needs $l_{\max} \sim 12/\sqrt{D_R\tau}$ terms.

(ii) Gaussian geometric Euler. $Q = \exp(\boldsymbol{\xi} \cdot \mathbf{L})$ with $\boldsymbol{\xi} \sim \mathcal{N}(0, 2D_R\tau \mathbb{K}_3)$. This is the obvious discretization of (5). Expanding χ_l to fourth order in $\omega = |\boldsymbol{\xi}|$, and using $\sum_m m^2 = \frac{1}{3}l(l+1)(2l+1)$, $\sum_m m^4 = \frac{1}{15}l(l+1)(2l+1)(3l^2+3l-1)$ and $\mathbb{E}|\boldsymbol{\xi}|^4 = 15s^4$, gives with $\Lambda = l(l+1)$

$$f_l^G = 1 - \Lambda D_R\tau + \frac{1}{6}\Lambda(3\Lambda-1)(D_R\tau)^2 + O(\tau^3) = e^{-\Lambda D_R\tau} - \frac{\Lambda}{6}(D_R\tau)^2 + O(\tau^3). \quad (9)$$

The one-step weak error is $O(\tau^2)$, so the global error after t/τ steps is $O(\tau)$. Equivalently, $f_l^G = e^{-\Lambda D_{\text{eff}}\tau} + O(\tau^3)$ with $D_{\text{eff}} = D_R(1 + D_R\tau/6)$: the Gaussian step over-diffuses, which is a curvature effect of $SO(3)$.

(iii) Curvature-corrected Gaussian. Because the error in (9) is linear in Λ , it can be absorbed into the variance for all l at once:

$$\boldsymbol{\xi} \sim \mathcal{N}(0, 2D_R\tau(1 - \frac{1}{6}D_R\tau)\mathbb{K}_3) \implies f_l = e^{-\Lambda D_R\tau} + O(\tau^3). \quad (10)$$

This is as cheap as (ii) and has one-step weak error $O(\tau^3)$.

Fig. 1 confirms these scalings. In panel (a) the one-step error of the geometric-Euler step scales as $(D_R\tau)^2$ and that of the corrected step as $(D_R\tau)^3$. In panel (b), Monte-Carlo (MC) orientational correlation functions over several coarse steps ($D_R\tau = 0.1$) show a linear-in- t drift for the geometric-Euler step, exactly as $(f_l^G/e^{-\Lambda D_R\tau})^{t/\tau}$ predicts. The exact and corrected samplers show no drift.

3 Route 1: piecewise-constant orientation

Algorithm. For each of N trajectories:

1. Draw R_0 from the Haar measure (the stationary state; we use normalized Gaussian quaternions).
2. For $k = 0, \dots, n-1$: propagate the spin with $U_k = e^{-i\tau H(R_k)}$, which is exact for one spin and costs one Rodrigues rotation of $\mathbf{M}$. Then update $R_{k+1} = R_k Q_k$.
3. Average the observables over trajectories.

A single trajectory $R(t)$ is only Hölder- $\frac{1}{2}$ continuous, so the strong (pathwise) error is $O(\tau^{1/2})$. Ensemble averages behave much better.

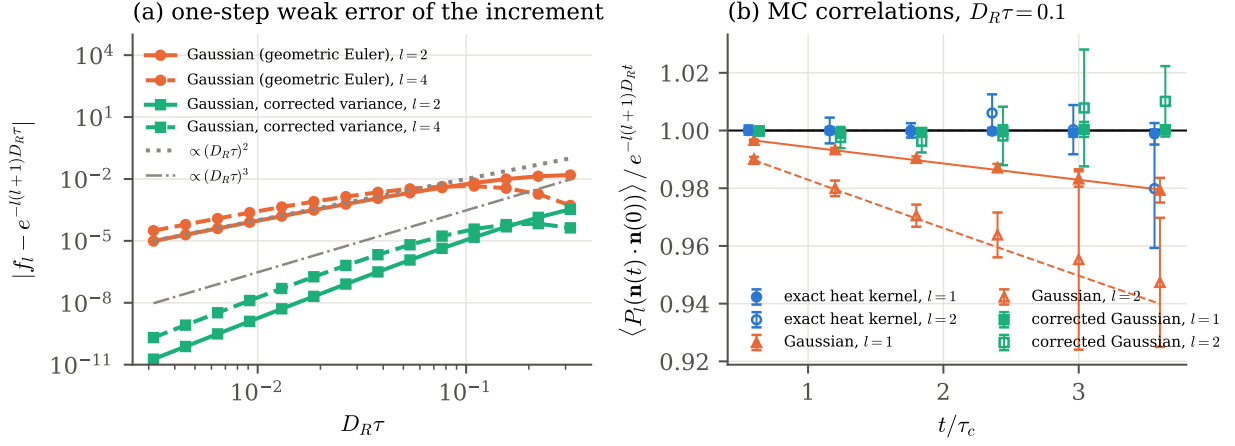


Figure 1: Increment samplers for the $SO(3)$ Wiener process. (a) One-step weak error $|f_l - e^{-l(l+1)D_R\tau}|$ from 1D quadrature of (7). (b) MC correlations $\langle P_l(\mathbf{n}(t) \cdot \mathbf{n}(0)) \rangle$ divided by the exact $e^{-l(l+1)D_R t}$; $4 \times 2.5 \times 10^5$ trajectories, error bars 2σ . Lines show $(f_l/e^{-\Delta D_R\tau})^{t/\tau}$.

Proposition 1 (Averaged Route 1 is a splitting of the stochastic Liouville equation). *Let $\mathcal{L}_H = -i[H(R), \cdot]$ act pointwise in R , and let $\mathcal{Q}$ be the Markov transfer operator of one increment. For the exact kernel, $\mathcal{Q} = e^{\tau D_R \nabla^2}$. Then*

$$\mathbb{E} \left[\prod_k U_k \rho_0 \prod_k U_k^\dagger \right] = \langle 1 | (\mathcal{Q} e^{\tau \mathcal{L}_H})^n | \rho_0 \otimes \text{Haar} \rangle, \quad (11)$$

where $\langle 1 |$ integrates over $SO(3)$. Since $\mathcal{Q}^{1/2} |\text{Haar}\rangle = |\text{Haar}\rangle$ and $\langle 1 | \mathcal{Q}^{1/2} = \langle 1 |$, this Lie product equals the Strang product $\langle 1 | (\mathcal{Q}^{1/2} e^{\tau \mathcal{L}_H} \mathcal{Q}^{1/2})^n | \cdot \rangle$. With exact or corrected-Gaussian increments the averaged scheme therefore has global weak error $O(\tau^2)$. With geometric-Euler increments, $\mathcal{Q}^G - e^{\tau D_R \nabla^2} = O(\tau^2)$, and the global error is $O(\tau)$.

The first identity follows from the Markov property: condition on R_k and integrate out R_{k+1} , one step at a time. For secular, axial CSA, $H = \delta\omega(n_z)I_z$ only couples $m=0$ functions of n_z . The averaged scheme then reduces to an even- l Legendre basis with $\mathcal{Q} = \text{diag}(f_l)$. This gives the *exact expectation* of Route 1, free of MC noise, and we use it to measure the Trotter error in Fig. 3.

Discrete-time correlations: why $\tau \lesssim \tau_c$ is needed. With exact increments, the sampled field $\mathbf{b}_k = \mathbf{b}(R_k)$ has $\langle b_{i,k} b_{i,k'} \rangle = C_{ii} r^{|k-k'|}$ with $r = e^{-\tau/\tau_c}$. Holding it constant on each step gives the phase-averaged spectral density

$$S_{ii}^{(\tau)}(\omega) = C_{ii} \tau \text{sinc}^2\left(\frac{\omega\tau}{2}\right) \frac{1-r^2}{1-2r \cos \omega\tau + r^2}, \quad \frac{S_{ii}^{(\tau)}(0)}{S_{ii}(0)} = \frac{\tau}{2\tau_c} \coth \frac{\tau}{2\tau_c} = 1 + \frac{\tau^2}{12\tau_c^2} + \dots \quad (12)$$

Here $S_{ii}(0) = 2C_{ii}\tau_c$ is the continuous-time value. To leading order in δ , Route 1 thus gives $R_1^{(\tau)} = \frac{1}{2}[S_{xx}^{(\tau)} + S_{yy}^{(\tau)}](\omega_0)$ and $R_2^{(\tau)} = \frac{1}{2}S_{zz}^{(\tau)}(0) + \frac{1}{2}R_1^{(\tau)}$. The relative error is $\tau^2/(12\tau_c^2)$, consistent with the proposition. It also exposes the failure mode: for $\tau \gg \tau_c$ one gets $S^{(\tau)}(0) \rightarrow C\tau$, so $R_2^{(\tau)} \propto \Gamma\tau/\tau_c$ (with $\Gamma = \delta^2\tau_c$) diverges instead of saturating.

4 The white-noise limit

Take $\tau_c \rightarrow 0$ at fixed $\Gamma = \delta^2\tau_c$, so that $\omega_0\tau_c \rightarrow 0$ and $\delta\tau_c \rightarrow 0$. Then $\mathbf{b}(t)$ becomes delta-correlated,

$$\langle b_i(t) b_j(t') \rangle \rightarrow 2\gamma_i \delta_{ij} \delta(t-t'), \quad \gamma_i = C_{ii}\tau_c = \Gamma \left(\frac{1}{15}, \frac{1}{15}, \frac{4}{45} \right), \quad (13)$$

and the orientation disappears from the problem. The right object to sample on a step $\tau \gg \tau_c$ is the *integrated field*, not the instantaneous one:

$$\mathbf{X}_k = \int_{t_k}^{t_k+\tau} \mathbf{b}(t) dt. \quad (14)$$

This is a sum of $\sim \tau/\tau_c$ weakly correlated contributions. By the central limit theorem it is Gaussian with $\text{Cov}(\mathbf{X}_k) = 2 \text{diag}(\gamma) \tau + O(\tau_c)$, and it is independent between steps. Its non-Gaussian cumulants vanish as

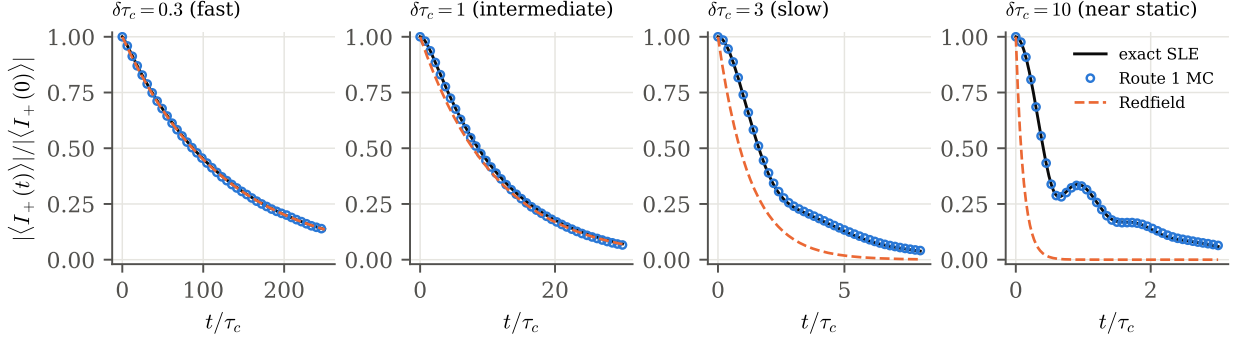


Figure 2: Secular axial CSA, $\tau_c = 1$: transverse coherence $|\langle I_+(t) \rangle|$ from Route 1 (exact increments, 4×10^4 trajectories, $\tau = \min(0.05, 0.05/\delta) \tau_c$) compared with the exact stochastic-Liouville solution (Legendre basis, $l_{\max} = 80$) and Redfield $e^{-R_2 t}$. The maximum deviation from the exact curve is $(4-8) \times 10^{-3}$ in every panel, at the MC noise level. Route 1 captures the slow-motion, non-exponential and near-static (powder-like) regimes, where Redfield theory fails.

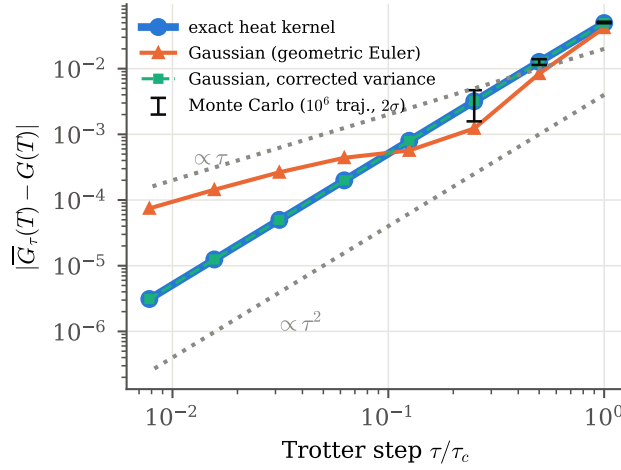


Figure 3: Weak Trotter error of Route 1 for secular CSA at $\delta\tau_c = 2$ and $T = 4\tau_c$. Curves show the exact expectation of the scheme (transfer-operator product), not MC estimates. The exact-kernel and corrected-Gaussian samplers converge as τ^2 and are indistinguishable. The geometric-Euler sampler crosses over to τ^1 once its $O(D_R^2 \tau)$ bias dominates. Black error bars show direct MC (10×10^5 trajectories); they agree with the transfer-operator prediction to within 2σ , confirming the proposition.

powers of $\delta\tau_c$. Route 1 thus becomes a Trotterized Stratonovich SDE, the Wong–Zakai limit of the colored noise:

$$d\psi = -i\left(\omega_0 I_z dt + \sum_i \sqrt{2\gamma_i} I_i \circ dW_i\right)\psi, \quad U_k = e^{-i\omega_0 I_z \tau/2} e^{-i\mathbf{X}_k \cdot \mathbf{I}} e^{-i\omega_0 I_z \tau/2}. \quad (15)$$

Its average is the Lindbladian $\mathcal{L}\rho = -i\omega_0 [I_z, \rho] - \sum_i \gamma_i [I_i, [I_i, \rho]]$, with $R_1 = \gamma_x + \gamma_y = \frac{2}{15}\Gamma$ and $R_2 = \gamma_z + \frac{1}{2}(\gamma_x + \gamma_y) = \frac{7}{45}\Gamma$. These are (4) at $\omega_0 \tau_c = 0$.

Remark 1 (Weak order of the kick step). *Averaging $e^{-iX_i I_i}$ over a single Gaussian component gives exactly $e^{-\tau\gamma_i [I_i, [I_i, \cdot]]}$. The joint kick $e^{-i\mathbf{X} \cdot \mathbf{I}}$ differs from $e^{\tau\mathcal{L}_{\text{noise}}}$ at $O(\gamma^2 \tau^2)$ because the components do not commute. A palindromic product of single-axis kicks, $e^{-i\omega_0 I_z \tau/2} K_x(\frac{\tau}{2}) K_y(\frac{\tau}{2}) K_z(\tau) K_y(\frac{\tau}{2}) K_x(\frac{\tau}{2}) e^{-i\omega_0 I_z \tau/2}$ with $K_i(s) = e^{-iX_i I_i}$ and $X_i \sim \mathcal{N}(0, 2\gamma_i s)$, averages to a Strang product of exact exponentials and so has weak order two. For the parameters here, $\gamma\tau \lesssim 10^{-2}$ and the difference is below the MC noise.*

Frequency-resolved kicks (finite $\omega_0 \tau_c$). In high-field liquid-state NMR, $\omega_0 \tau_c$ is often $O(10^{-1}-1)$ while $\delta\tau_c \ll 1$. One can then coarse-grain on $\tau_c \ll \tau \ll 1/\max(R_1, R_2)$ in the Zeeman interaction frame. The transverse integrated field $X_+ = \int b_+(t) e^{i\omega_0 t} dt$ is circular Gaussian: $\mathbb{E}X_+^2 = 0$ by isotropy, and $\mathbb{E}|X_+|^2 = 2 \cdot 2\gamma_\perp(\omega_0)\tau$ with $\gamma_\perp(\omega_0) = C_{xx}\tau_c/(1 + \omega_0^2 \tau_c^2)$. The longitudinal field keeps $\gamma_z = C_{zz}\tau_c$. Kicks in the rotating frame with covariance $2 \text{diag}(\gamma_\perp, \gamma_\perp, \gamma_z)\tau$ reproduce (4) for every $\omega_0 \tau_c$. They do not capture the second-order

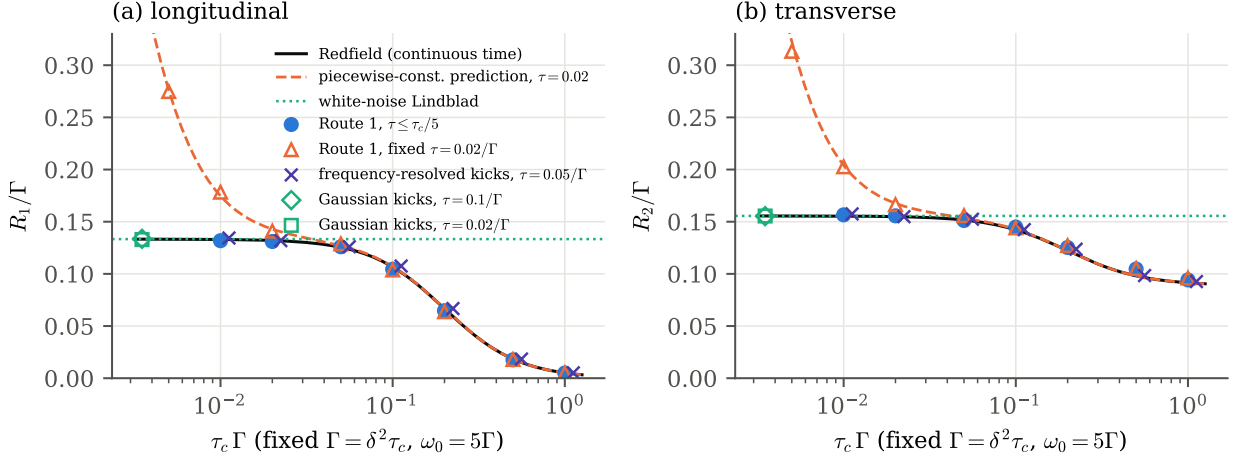


Figure 4: Approach to the white-noise limit for full (non-secular) CSA at fixed $\Gamma = \delta^2 \tau_c$ and $\omega_0 = 5\Gamma$. Solid: Redfield (4). Dotted: white-noise Lindblad. Dashed: prediction (12) for the piecewise-constant scheme at fixed $\tau = 0.02/\Gamma$. The white-noise Gaussian-kick results (diamond, square) represent $\tau_c \rightarrow 0$ and are drawn at the left edge. Route 1 uses 2×10^4 trajectories; the kick schemes use 10^5 – 2×10^5 .

dynamic frequency shift, which is a deterministic $O(\delta^2 \tau_c \cdot \omega_0 \tau_c)$ correction that can be added to the drift. This scheme is the stochastic unraveling of Redfield theory. It is valid when $\delta \tau_c \ll 1$, whereas the full Route 1 of Section 3 also covers slow motion.

Numerics (Fig. 4). We fix $\Gamma = 1$ and $\omega_0 = 5\Gamma$, vary τ_c , and extract $R_{1,2}$ from exponential fits on $t \in [1, 10]/\Gamma$:

- *Route 1 with $\tau \leq \tau_c/5$* follows Redfield across the crossover $\omega_0 \tau_c \sim 1$ and into the white-noise plateau.
- *Route 1 with fixed $\tau = 0.02/\Gamma$* is accurate while $\tau \lesssim \tau_c$, and blows up for $\tau_c \ll \tau$, exactly as the discrete spectral density (12) predicts (dashed line, no free parameters).
- *Gaussian kicks* (15) give $R_1 = 0.1336$ and 0.1329 at $\tau = 0.1$ and 0.02 (2×10^5 trajectories), against the Lindblad value $2/15 = 0.1333$. They give $R_2 = 0.1553$ and 0.1555 , against $7/45 = 0.1556$. These results need no orientations and allow steps $\tau \gg \tau_c$.
- *Frequency-resolved kicks* ($\times$) lie on the Redfield curve for all τ_c .

5 Two spins: dipolar cross relaxation and DD/CSA interference

5.1 Model: several interactions, one orientation

Consider a rigid heteronuclear pair I, S at fixed distance, carrying the dipolar (DD) coupling and a CSA on each spin. Every interaction is a traceless symmetric tensor T_λ , fixed in the molecular frame, contracted with a spin operator:

$$H(R) = \omega_I I_z + \omega_S S_z + \sum_\lambda \sum_{ij} (RT_\lambda R^\top)_{ij} O_{ij}^\lambda, \quad \lambda \in \{\text{DD}, \text{CSA}_I, \text{CSA}_S\}. \quad (16)$$

With the bond along the molecular z axis:

- **Dipolar coupling:** $T_{\text{DD}} = -3d(\hat{z}\hat{z}^\top - \frac{1}{3}\mathbb{I})$ and $O_{ij}^{\text{DD}} = I_i S_j$, which gives $H_{\text{DD}} = d[\mathbf{I} \cdot \mathbf{S} - 3(\mathbf{I} \cdot \mathbf{n})(\mathbf{S} \cdot \mathbf{n})]$ with $d = \mu_0 \hbar \gamma_I \gamma_S / (4\pi r^3)$.
- **CSA on spin I :** $T_{\text{CSA}, I} = \delta_I R_y(\beta_I) A R_y(\beta_I)^\top$, with A from (1) and $O_{ij} = \delta_{iz} I_j$, as in Section 1. The unique CSA axis is tilted by β_I from the bond. The CSA on spin S is defined in the same way.

The physics of cross-correlation is entirely in the fact that the same $R(t)$ appears in every term of (16). Everything below follows from this one statement.

One rank-2 random field. Let $\{E_a\}_{a=1}^5$ be a real orthonormal basis of symmetric traceless 3×3 matrices, $\text{tr}(E_a E_b) = \delta_{ab}$. Define

$$\mathcal{D}_{ab}(R) = \text{tr}(E_a R E_b R^\top), \quad t_{\lambda,b} = \text{tr}(E_b T_\lambda), \quad Q_a^\lambda = \sum_{ij} (E_a)_{ij} O_{ij}^\lambda. \quad (17)$$

Here $\mathcal{D}(R)$ is the real form of the Wigner matrix $D^{(2)}(R)$. The fluctuating Hamiltonian then reads

$$H_1(R) = \sum_{a,b=1}^5 \mathcal{D}_{ab}(R) W_{ab}, \quad W_{ab} = \sum_{\lambda} t_{\lambda,b} Q_a^\lambda. \quad (18)$$

The index b labels molecular-frame tensor components and a labels lab-frame components. Under isotropic diffusion, orthogonality of the Wigner functions gives

$$\mathbb{E}[\mathcal{D}_{ab}(R(0)) \mathcal{D}_{cd}(R(t))] = \frac{1}{5} \delta_{ac} \delta_{bd} e^{-|t|/\tau_c}. \quad (19)$$

The correlation between two interactions is therefore set by the overlap of their molecular-frame tensors, $\sum_b t_{\lambda,b} t_{\lambda',b} = \text{tr}(T_\lambda T_{\lambda'})$. For DD and an axial CSA this overlap is

$$\text{tr}(T_{\text{DD}} T_{\text{CSA},I}) = -2 d \delta_I P_2(\cos \beta_I), \quad (20)$$

which vanishes at the magic angle $\beta_I = 54.74^\circ$.

Redfield reference. Split each operator into Zeeman frequency components, $W_{ab} = \sum_{\nu} W_{ab}[\nu]$, where $(W[\nu])_{pq} \neq 0$ only if $E_p - E_q = \nu$. Bloch–Redfield theory with (19) gives

$$\dot{\rho} = -i[H_0, \rho] - \frac{1}{5} \sum_{ab} \sum_{\nu} k(\nu) [W_{ab}, [W_{ab}[\nu], \rho]], \quad k(\nu) = \frac{\tau_c}{1 + i\nu\tau_c}. \quad (21)$$

The secular version keeps only $\text{Re } k = \frac{1}{2} \tilde{J}(\nu)$, with $\tilde{J}(\nu) = 2\tau_c/(1 + \nu^2\tau_c^2)$, and pairs $W[\nu]$ with $W[\nu]^\dagger$. We verified numerically that (21) reproduces the Solomon rates $R_1^I = \frac{d^2}{4} [J(\omega_I - \omega_S) + 3J(\omega_I) + 6J(\omega_I + \omega_S)] + \frac{1}{3} \delta_I^2 J(\omega_I)$ and $\sigma_{IS} = \frac{d^2}{4} [6J(\omega_I + \omega_S) - J(\omega_I - \omega_S)]$ exactly. It also reproduces the DD/CSA cross-correlated rates

$$\eta_z = -d \delta_I P_2(\cos \beta_I) J(\omega_I), \quad \eta_{xy} = -\frac{1}{6} d \delta_I P_2(\cos \beta_I) [4J(0) + 3J(\omega_I)], \quad (22)$$

to machine precision for three random parameter sets. These are defined by $\frac{d}{dt} c_{2I_z S_z} = -\eta_z c_{I_z} + \dots$ and $\frac{d}{dt} c_{2I_+ S_z} = -\eta_{xy} c_{I_+} + \dots$, with $c_O = \text{tr}(O\rho)/\text{tr}(O^\dagger O)$. The magnitudes agree with the standard results; the signs follow our conventions for d and δ_I . The cross terms are exactly the $\lambda \neq \lambda'$ contributions to $W_{ab} \otimes W_{ab}$. Replacing $W_{ab} \otimes W_{ab}$ by $\sum_{\lambda} W_{ab}^\lambda \otimes W_{ab}^\lambda$ (one independent orientation per interaction) leaves R_1 and σ_{IS} unchanged but sets $\eta_z = \eta_{xy} = 0$.

5.2 Route 1 for the pair

Route 1 carries over unchanged. Each trajectory carries one R_k , updated with exact heat-kernel increments. The 4×4 Hamiltonian $H(R_k)$ of (16) is exponentiated exactly (batched `expm`), and $\rho \mapsto U_k \rho U_k^\dagger$. The correlations are correct by construction, because every interaction is evaluated at the same R_k . The instructive *wrong* implementation draws an independent $R_k^{(\lambda)}$ for each interaction. It reproduces all auto-relaxation and dipolar cross relaxation, and misses every DD/CSA interference effect.

5.3 Gaussian kicks with the correct correlations

In the white-noise regime of Section 4 we again replace the orientation by the integrated field. Now, however, the integrated field is the whole 5×5 matrix

$$\Xi_{ab} = \int_{t_k}^{t_k+\tau} \mathcal{D}_{ab}(R(t)) dt, \quad \Xi_{ab} \text{ i.i.d. } \mathcal{N}(0, \tilde{J}(0) \tau/5), \quad Y_k = \sum_{ab} \Xi_{ab} W_{ab}, \quad U_k = e^{-iY_k}. \quad (23)$$

One Ξ is shared by all interactions. Written per interaction, the kick tensor of λ is $K_\lambda = \sum_a (\Xi t_\lambda)_a E_a$: every interaction receives a different linear image of the *same* Gaussian matrix. This is the Gaussian surrogate of “integrate $RT_\lambda R^\top$ along one trajectory”. For a single spin (Section 4), Ξ collapses onto the three field components used there. The covariance of all kick components has rank at most $5 \times \text{rank}\{t_\lambda\} \leq 25$. For coaxial axial tensors ($\beta_I = \beta_S = 0$) the rank is only 5, since only $\Xi \hat{t}$ enters.

Frequency-resolved kicks. Different spectral components must be kept apart when $\omega\tau_c$ is not small. Work in the Zeeman interaction frame and draw an independent $\Xi^{(\nu)}$ for each distinct frequency $\nu \geq 0$, *shared by all interactions*:

$$Y_k = \sum_{ab} \left(\Xi_{ab}^{(0)} W_{ab}[0] + \sum_{\nu>0} [\Xi_{ab}^{(\nu)} W_{ab}[\nu] + \text{h.c.}] \right), \quad \mathbb{E} |\Xi_{ab}^{(\nu)}|^2 = \tilde{J}(\nu) \tau / 5. \quad (24)$$

Here $\Xi^{(\nu>0)}$ is circular complex Gaussian and $\Xi^{(0)}$ is real. The average of $e^{-iY} \rho e^{iY}$ is $\rho + \tau \mathcal{R}_{\text{sec}} \rho + O(\tau^2)$, where $\mathcal{R}_{\text{sec}}$ is the secular part of (21). This includes all auto- and cross-correlated terms, each at its correct $J(\nu)$. For a heteronuclear pair the relevant frequencies are $0, \omega_S, \omega_I - \omega_S, \omega_I, \omega_I + \omega_S$. The secular approximation needs $\tau \gg 1 / \min |\nu - \nu'|$, which is trivially satisfied at high field.

Two tempting simplifications are wrong:

- Drawing an independent $\Xi^{(\lambda)}$ per interaction removes the DD/CSA interference.
- Drawing independent kicks per spin operator (for example per Pauli string) discards the off-diagonal covariance between operators driven by the same $\mathcal{D}_{ab}$.

The correct covariance is never diagonal in the interaction or operator basis. It is always diagonal in (a, b, ν) .

5.4 Numerical examples

We take $\tau_c = 1$, $d = \delta_I = 0.25$, $\delta_S = 0.1$, $\omega_I = 1$, $\omega_S = 0.4$, with coaxial CSA ($\beta_I = \beta_S = 0$) unless stated otherwise. Route 1 uses $\tau = 0.1\tau_c$ and 4×10^4 trajectories. The frequency-resolved kicks use $\tau = \tau_c$ and 10^5 trajectories.

- **Longitudinal (Fig. 5).** Start from $\rho(0) \propto I_z$. The $I_z \rightarrow S_z$ transfer (the NOE, σ_{IS}) is the same with shared and with independent orientations. The $I_z \rightarrow 2I_z S_z$ transfer is purely DD/CSA interference. It reaches 0.14 for the shared orientation and is zero (kicks) or near zero (Route 1; see below) for independent orientations. The correlated kicks follow Redfield within MC error: at $t = 20\tau_c$, $\langle 2I_z S_z \rangle = 0.1309$ against 0.1301 (secular Redfield) and 0.1320 (non-secular).
- **Transverse (Fig. 6).** Start from $\rho(0) \propto I_x$. The two doublet components $I_+ (\frac{1}{2} \pm S_z)$ relax at different rates only with a shared orientation. This is the TROSY effect, generated here by the correlated kicks or by Route 1 without any extra input.
- **Angular dependence (Fig. 7a,b).** Both the longitudinal transfer and the differential transverse rate follow $P_2(\cos \beta_I)$ and change sign at the magic angle. The kicks track Redfield at every angle; Route 1 is shown at $0^\circ, 54.7^\circ$ and 90° .

Why Route 1 differs from Redfield by $\sim 10\%$: non-Gaussian orientation noise. Route 1 lies systematically below Redfield for $\langle 2I_z S_z \rangle$ and $|\langle S_z \rangle|$ (Fig. 5). This is not an initial-slip effect: the time-convolutionless second-order (TCL2) equation, which includes the slip, differs from Redfield by less than 1%. It is not Monte-Carlo noise either. The deviation is *odd* in the couplings. Flipping the sign of all of d, δ_I, δ_S flips the sign of the deviation, and it grows linearly with the coupling scale: $\approx 2.5\%$ at $d\tau_c = 0.05$ and $\approx 12.5\%$ at $d\tau_c = 0.25$ (Fig. 7c). The sign-symmetrized Route 1 average agrees with TCL2 to $\leq 0.6\%$ at all coupling strengths.

The origin is the third cumulant of the orientation noise. The functions $\mathcal{D}_{ab}(R)$ are bounded and non-Gaussian, and $\mathbb{E}[\mathcal{DDD}] \neq 0$ because $2 \otimes 2 \otimes 2 \ni 0$. For example, $\langle P_2^3 \rangle = 2/35$, and the corresponding Gaunt coefficient is nonzero. Third-order terms in the cumulant expansion therefore contribute at relative order $\|H_1\| \tau_c$. The same mechanism explains the small $I_z \rightarrow 2I_z S_z$ transfer in the *independent*-orientation Route 1 run: it is dominated by $\mathbb{E}[F_{\text{DD}}^3] \neq 0$, which needs no correlation between interactions. Gaussian kicks have no third cumulant and reproduce second-order (Redfield) theory only.

For liquid-state NMR, $\|H_1\| \tau_c \sim 10^{-5}$, so these corrections are utterly negligible. There the correlated Gaussian kicks are the right tool, and Route 1 is the reference that shows why.

6 Practical recipe and cost

1. **Sampling the $SO(3)$ Wiener process.** Use exact heat-kernel increments (tabulated inverse CDF), or the Gaussian with variance $2D_{RT}(1 - D_{RT}/6)$. Both give weak order two in the averaged Trotter scheme; the naive Gaussian gives order one. Start from the Haar measure, so that the Lie ordering is automatically equivalent to Strang.

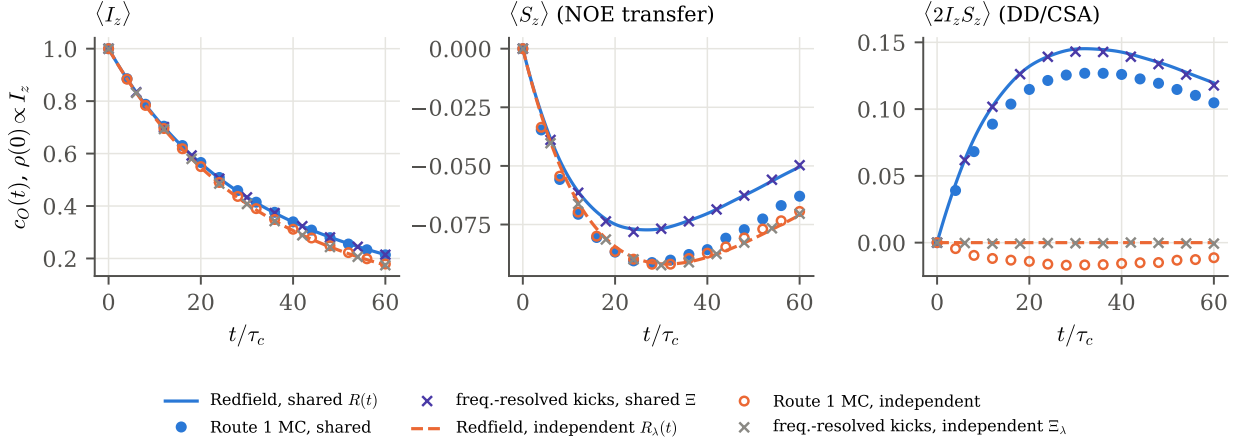


Figure 5: Longitudinal two-spin dynamics from $\rho(0) \propto I_z$. Lines: Bloch–Redfield (21) with one shared orientation (solid) or independent orientations per interaction (dashed). Markers: Route 1 MC and frequency-resolved kicks (24), with shared or independent orientations/fields. The NOE transfer to S_z is insensitive to the correlations. The $2I_z S_z$ transfer exists only with a shared orientation.

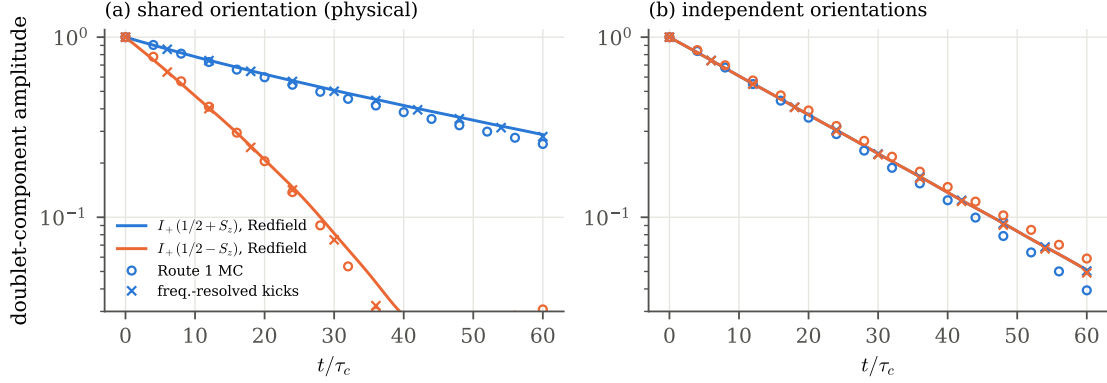


Figure 6: Doublet components $|c_{I_+} \pm c_{2I_+ S_z}|$ from $\rho(0) \propto I_x$. (a) With a shared orientation, DD/CSA interference splits the two lines into a slowly relaxing (TROSY) component and a fast one. (b) With independent orientations the splitting disappears; the residual difference in the Route 1 markers is the odd-order effect discussed in the text.

- Step size.** Piecewise-constant orientation needs $\tau \lesssim \tau_c$ (relative rate error $\tau^2/12\tau_c^2$) and $\|H\|\tau \lesssim 0.1$. Exact increments remove any extra condition on $D_R\tau$ from the orientation dynamics itself.
- White-noise regime** ($\delta\tau_c \ll 1$, $\tau \gg \tau_c$). Do not sample orientations. Sample Gaussian integrated fields with covariance $2\gamma\tau$, frequency-resolved if $\omega_0\tau_c$ is not small, and Trotterize the resulting Stratonovich SDE palindromically.
- Several interactions (Section 5).** Write every interaction as $\sum_{ab} \mathcal{D}_{ab}(R)W_{ab}$. In Route 1, evaluate all interactions at the same R_k . In the kick scheme, draw one 5×5 Gaussian $\Xi^{(\nu)}$ per Zeeman frequency and share it across all interactions. Never draw noise independently per interaction or per spin operator.

Why item 3 matters for liquid-state NMR. For a ^{13}C carbonyl at 14.1 T ($\omega_0/2\pi \approx 150$ MHz, $\Delta\sigma \approx 150$ ppm) we have $\delta \approx 1.4 \times 10^5 \text{ s}^{-1}$. With $\tau_c = 100$ ps this gives $\delta\tau_c \approx 10^{-5}$, $\Gamma \approx 2 \text{ s}^{-1}$ and $\omega_0\tau_c \approx 0.1$. Direct Route 1 with $\tau \leq \tau_c/5$ over a 1 s experiment would need $\sim 5 \times 10^{10}$ steps per trajectory. The kick scheme needs $\tau \sim 10^{-3} - 10^{-2}$ s, i.e. $\sim 10^2 - 10^3$ steps. For many spins, the kicks for different interactions (CSA, dipolar, and their cross-correlations) are correlated Gaussians whose joint covariance is the matrix of spectral densities $J_{\alpha\beta}(\omega)$ of the underlying orientation process. This is the natural bridge to the spatially-correlated-noise models in the rest of this project.

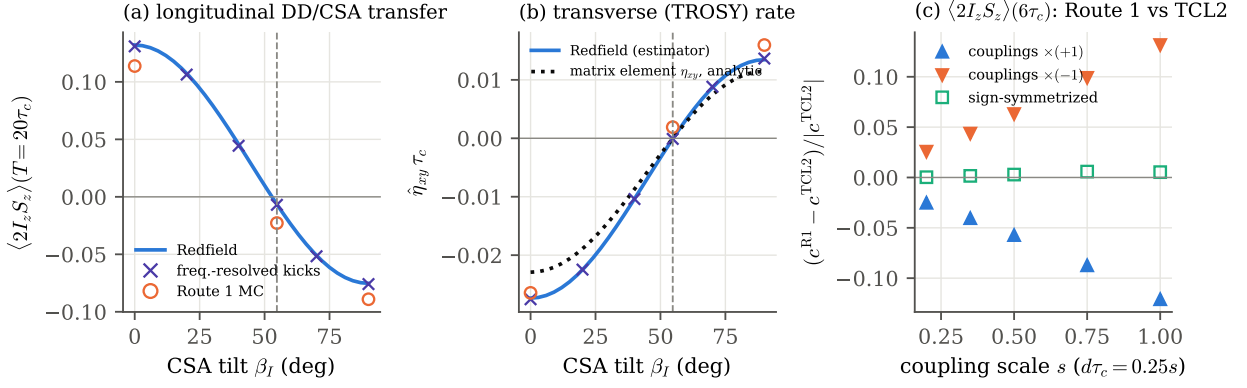


Figure 7: (a) $\langle 2I_z S_z \rangle$ at $T = 20\tau_c$ and (b) the TROSY estimator $\hat{\eta}_{xy} = \ln(a_-/a_+)/2T$ versus CSA tilt β_I . The dotted line in (b) is the matrix element (22). The estimator differs from it slightly because in-phase and antiphase coherences have different auto-relaxation rates. The dashed vertical line marks the magic angle. (c) Relative deviation of Route 1 from TCL2 for $\langle 2I_z S_z \rangle(6\tau_c)$ as the couplings are scaled by s , for both overall signs (2×10^5 trajectories each). The deviation is odd in the couplings and linear in s , and it cancels in the sign-symmetrized average.

A Code

All figures are generated by `rotdiff.py`, `pair.py` (two-spin model, Route 1, kicks, Redfield and TCL2 references) and the `fig*.py/run_pair.py/plot*.py` scripts that accompany this note. The core pieces are:

```
def heat_kernel_cdf(Dt, ngrid=4000):
    lmax = int(12/np.sqrt(Dt)) + 30
    w = np.linspace(0, np.pi, ngrid); cdf = 0.5*(w - np.sin(w))
    for l in range(1, lmax+1):
        cdf += (2*l+1)*np.exp(-l*(l+1)*Dt)*0.5*(np.sin(l*w)/l - np.sin((l+1)*w)/(l+1))
    cdf = np.maximum.accumulate(np.clip(2/np.pi*cdf, 0, None)); return w, cdf/cdf[-1]

def increments(N, D, tau, kind): # rotation vectors xi (axis*angle)
    if kind == "exact":
        ang = np.interp(rng.random(N), cdf, w)
        ax = rng.normal(size=(N,3)); ax /= np.linalg.norm(ax, axis=1, keepdims=True)
        return ax*ang[:,None]
    s2 = 2*D*tau*(1 - D*tau/6 if kind == "gauss2" else 1)
    return rng.normal(scale=np.sqrt(s2), size=(N,3))

# Route 1 step: spin first (orientation constant on the step), then orientation
h = csa_field(R, delta, w0) # b(R) + w0 zhat, eq. (1)
M = rotate_bloch(M, h, tau) # exact exp(-i tau h.I) on Bloch vector
R = R @ rodrigues(increments(N, D, tau, kind))

# White-noise limit: Gaussian kick of the integrated field, Strang with Zeeman
X = rng.normal(size=(N,3))*np.sqrt(2*gamma*tau)
M = rot_z(M, w0*tau/2); M = rotate_bloch(M, X, 1.0); M = rot_z(M, w0*tau/2)

# Two spins (pair.py): W[a,b] = sum_lam t_lam[b] Q_lam[a], split by Zeeman frequency nu.
# ONE Xi per frequency, shared by DD, CSA_I, CSA_S -> correct cross-correlations.
Y = 0
for nu in nus: # distinct nu >= 0
    s2 = Jt(nu, tc)*tau/5 # Jt = 2 tc/(1+nu^2 tc^2)
    Xi = rng.normal(size=(N,5,5))*np.sqrt(s2) if nu == 0 else \
        (rng.normal(size=(N,5,5)) + 1j*rng.normal(size=(N,5,5)))*np.sqrt(s2/2)
    A = (Xi.reshape(N,25) @ W_nu[nu].reshape(25,16)).reshape(N,4,4)
    Y = Y + (A if nu == 0 else A + A.conj().transpose(0,2,1))
U = batch_expm_herm(Y, 1.0); rho = U @ rho @ U.conj().transpose(0,2,1)
```